HopRank: Hop-Aware Reranking for Dense Multi-Hop Retrieval

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Abstract

We present HopRank, which extends DenseHop with a hop-aware reranker that re-scores candidate passage chains using cross-attention between hops. On KaveriHop, HopRank reaches 53.4 EM on 2-hop and 36.0 EM on 3-hop questions, adding only 30 ms of latency per query. Hop-aware reranking closes the gap to graph-based retrieval on 2-hop questions but not on 3-hop questions.

Keywords: reranking, dense retrieval, multi-hop retrieval

1 Introduction

DenseHop [1] scores each hop independently, so an early mistake cannot be undone. We keep the DenseHop retriever and add a second stage, which we call hop-aware re-scoring, that judges whole chains rather than single passages.

2 Method

HopRank builds on DenseHop [1]: we take its top 50 candidate chains and re-score each chain with a cross-encoder that attends across all passages in the chain. The reranker is trained with chain-level hard negatives. Like most dense systems it inherits encoders from dense passage retrieval [4].

3 Experiments

We evaluate on KaveriHop [2] and compare against DenseHop and the graph-based GraphWalker-QA [3], quoting their published numbers. We also measure latency on a single GPU.

Method	Source	KaveriHop 2-hop EM	KaveriHop 3-hop EM	KaveriHop latency (ms/query)
DenseHop	[2]	48.2	31.5	–
DenseHop	ours	–	–	150
GraphWalker-QA	[3]	50.1	39.8	–
HopRank	ours	53.4	36.0	180

Table 1: Exact match and latency on KaveriHop.

4 Discussion

HopRank surpasses GraphWalker-QA by 3.3 EM on 2-hop questions but trails it by 3.8 EM on 3-hop questions. Hop-aware reranking closes the gap to graph-based retrieval on 2-hop questions but not on 3-hop questions.

5 Conclusion

Chain-level reranking is a cheap upgrade for dense multi-hop retrievers.

References

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